Laurent Martini-Bonnet

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1 Profile Summary

Master's student in Applied Mathematics at ETH Zurich, with a strong background and interest in machine learning and quantitative finance.

2 Skills

- **Programming Language and tools:** Python (Numpy, Pandas, TensorFlow, PyTorch), LaTeX
- **Languages:** French – native — English – C1

3 Education

3.1 M.S in Applied Mathematics — ETH — Zurich, ZH — 09/2024 - 08/2026

- Current GPA: 5.47/6
- Brownian Motion and Stochastic Calculus, Mathematical finance, Probabilistic AI, Large Language Models, Diffusion models, Machine learning for finance and complex systems

3.2 Exchange year — École polytechnique — Palaiseau, FR — 08/2023 - 07/2024

- Markov chains and martingales, Monte Carlo methods, Time series, Advanced probability topics, Optimization and control, Functional analysis

3.3 B.S. in Mathematics — EPFL — Lausanne, VD — 09/2021 - 07/2024

- Multivariate calculus, Algebra, Fourier analysis, Probability and measure theory, Statistics, Numerical analysis

4 Projects

- **Financial Time Series Generation via Schrödinger Bridges** (Present) - *Diffusion Models, Optimal Transport*: Exploring the intersection of generative models and Optimal Transport to develop and enhance robust synthetic market data generation.
- **Arbitrage-Free Neural SDE Model for Option Pricing**(02/2026) — *Stochastic calculus, PCA methods* : Theoretical audit of neural-SDE architectures. Streamlined no-arbitrage proofs. Proposed a novel theoretical approach to decode factors from option prices.
- **Local Risk Bounds for Statistical Aggregation** (01/2026) — *Aggregation, Theoretical machine learning* : Studied excess risk bounds from Q-aggregation and Exponential Weights methods. Implemented Exponential Weights aggregation procedure using Langevin dynamics approach.
- **Mixture Models for Bitcoin Volatility Forecasting** (06/2025) — *Python, Machine Learning* : Compared orderbook and sentiment feature-based models (including LSTM, XGBoost) with volatility-only models (including HAR,GARCH)

5 Extracurricular activities

Certificates in music theory and French horn. Concerts in Paris philarmonie, Versailles palace, Japan.